
SQUARES RISK

RESEARCH NOTES IN THE ENEXA AND QROM PROJECTS

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We reformulate results in Goessmann (2021) based on the formalism in Goessmann et al. (2026).

1 Quadratic exponential models

We consider distributions of variables $X_{[d]}$ and Y indexed by vector parameters $\tilde{\theta}[L] \in \mathbb{R}^d$ with L

$$\mathbb{P}^{\tilde{\theta}}[X_{[d]} = x_{[d]}, Y = y] = \frac{1}{\mathcal{Z}(\tilde{\theta})} \cdot \exp \left[-(\langle \tilde{\theta}[L], x_{[L]} \rangle_{[\emptyset]} - y)^2 \right].$$

Here we treat the index tuple $x_{[d]}$ as a vector $x[L]$.

Example 1 (Gaussian processes). *When $X_{[d]}$ and Y are continuous real variables, and the contraction is defined as Lebesgue integral, then $\mathbb{P}^{\tilde{\theta}}$ is a Gaussian process.*

Here we allow for arbitrary state sets to $X_{[d]}$ and Y , but are specially interested in discrete cases (where the contraction is the sum).

1.1 Cross entropy

The cross entropy between an empirical distribution $\mathbb{P}^D$ to the data $D = \left((x_0^j, \dots, x_{d-1}^j) : j \in [m] \right)$ and the model $\mathbb{P}^{\tilde{\theta}}$ is

$$\begin{aligned} \mathbb{H}[\mathbb{P}^D, \mathbb{P}^{\tilde{\theta}}] &= \frac{1}{m} \sum_{j \in [m]} -\ln \left[\mathbb{P}^{\tilde{\theta}}[X_{[d]} = x_{[d]}^j, Y = y^j] \right] \\ &= \ln[\mathcal{Z}(\tilde{\theta})] - \frac{1}{m} \sum_{j \in [m]} (\langle \tilde{\theta}[L], x^j \rangle_{[\emptyset]} - y^j)^2. \end{aligned}$$

The cross entropy differs by the term $\ln[\mathcal{Z}(\tilde{\theta})]$ from the squares risk

$$\frac{1}{m} \sum_{j \in [m]} \left(\langle \tilde{\theta}[L], x^j \rangle_{[\emptyset]} - y^j \right)^2.$$

1.2 Conditioned likelihood

When Y is real valued, then the distribution of Y conditioned on $X_{[d]} = x_{[d]}$ is

$$\mathbb{P}^{\tilde{\theta}}[Y = y | X_{[d]} = x_{[d]}],$$

that is a normal distribution with $\sigma = \sqrt{\frac{1}{2}}$.

The log likelihood of the data D conditioned on the variables $X_{[d]}$ is

$$\ln [\sqrt{\pi}] + \frac{1}{m} \sum_{j \in [m]} \left(\left\langle \tilde{\theta}[L], x^j \right\rangle_{[\emptyset]} - y^j \right)^2.$$

Thus, the difference to the squares risk is a constant term $\ln [\sqrt{\pi}]$, which can be ignored when optimizing over $\tilde{\theta}$.

1.3 Exponential family

By factorization of the squares term we get

$$\mathbb{P}^{\tilde{\theta}} [X_{[d]} = x_{[d]}, Y = y] = \frac{1}{\mathcal{Z}(\tilde{\theta})} \exp \left[- \left\langle \tilde{\theta}[L], x[L] \right\rangle_{[\emptyset]}^2 + 2y \left\langle \tilde{\theta}[L], x[L] \right\rangle_{[\emptyset]} - y^2 \right].$$

This is a member of the exponential family with the statistic

$$t(x, y) = (-x[L] \otimes x[\tilde{L}], 2y \cdot x[L], -y^2)$$

and the canonical parameter

$$\theta(\tilde{\theta}) = (\tilde{\theta}[L] \otimes \tilde{\theta}[\tilde{L}], \tilde{\theta}[L], 1).$$

2 Inference

The special case of Gaussian processes enable training methods like alternating least squares, see Goessmann (2021). They require sparsity assumptions by restricting $\tilde{\theta}$ to admit tensor network decompositions.

3 Probabilistic guarantees

From the basic inequality

$$\mathbb{H} \left[\mathbb{P}^D, \mathbb{P}^{\hat{\tilde{\theta}}} \right] \leq \mathbb{H} \left[\mathbb{P}^D, \mathbb{P}^{\tilde{\theta}^*} \right]$$

we derive an inequality consisting product and multiplier processes, scaling with $\|\hat{\tilde{\theta}} - \tilde{\theta}^*\|^2$ and $\|\hat{\tilde{\theta}} - \tilde{\theta}^*\|$. In combination with uniform concentration inequalities (involving e.g. Mendelson's Lambda functional), we then get a bound on $\|\hat{\tilde{\theta}} - \tilde{\theta}^*\|$.

References

- Alex Goessmann. *Uniform Concentration of Tensor and Neural Networks: An Approach towards Recovery Guarantees*. PhD Thesis, Technische Universität Berlin, Berlin, 2021. URL <https://depositonce.tu-berlin.de/handle/11303/15990>.
- Alex Goessmann, Janina Schütte, Maximilian Fröhlich, and Martin Eigel. A tensor network formalism for neuro-symbolic AI, January 2026. URL <http://arxiv.org/abs/2601.15442>. arXiv:2601.15442 [cs].